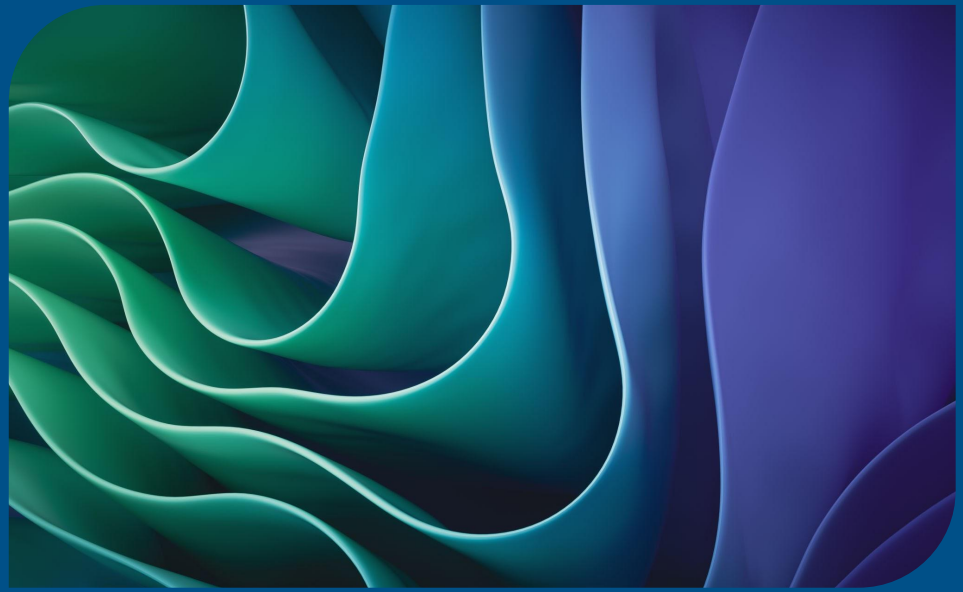
Super-Convergence: Very Fast Training of Neural Networks Using Large Learning Rates

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Presentation by Amir Bikineev



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2. Method
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4. Conclusions

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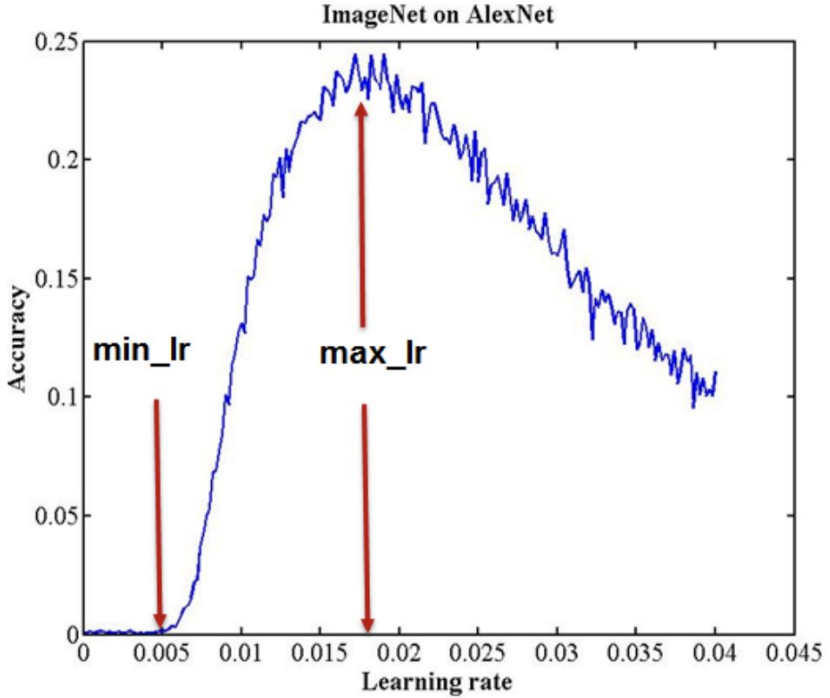
“Super-convergence” is a phenomenon where neural networks can be trained an order of magnitude faster than with standard training methods.

Super-convergence provides a greater boost in performance relative to standard training when the amount of labeled training data is limited.

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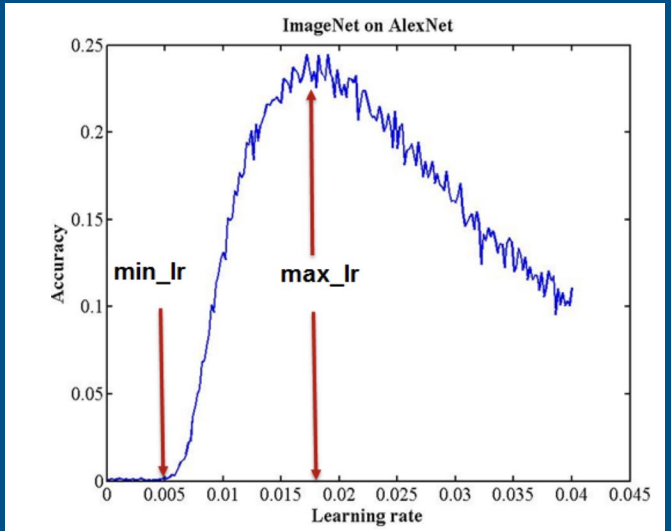
When super-convergence can be applicable?

To understand whether we can train model with super-convergence technique, we need first to make LR range test



(a) Typical learning rate range test result where there is a peak to indicate max_lr.

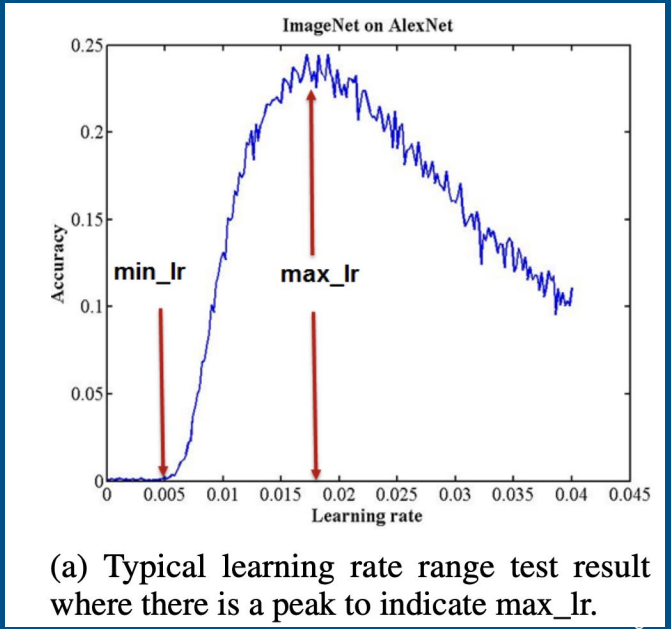
In the LR range test, training starts with a zero or very small learning rate which is slowly increased linearly throughout a pre-training run. This provides information on how well the network can be trained over a range of learning rates.



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When starting with a small learning rate, the network begins to converge and, as the learning rate increases, it eventually becomes too large and causes the training/test accuracy to decrease.

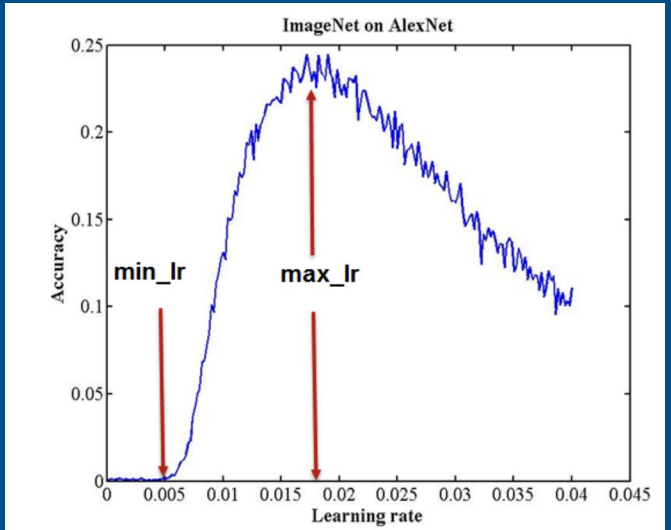


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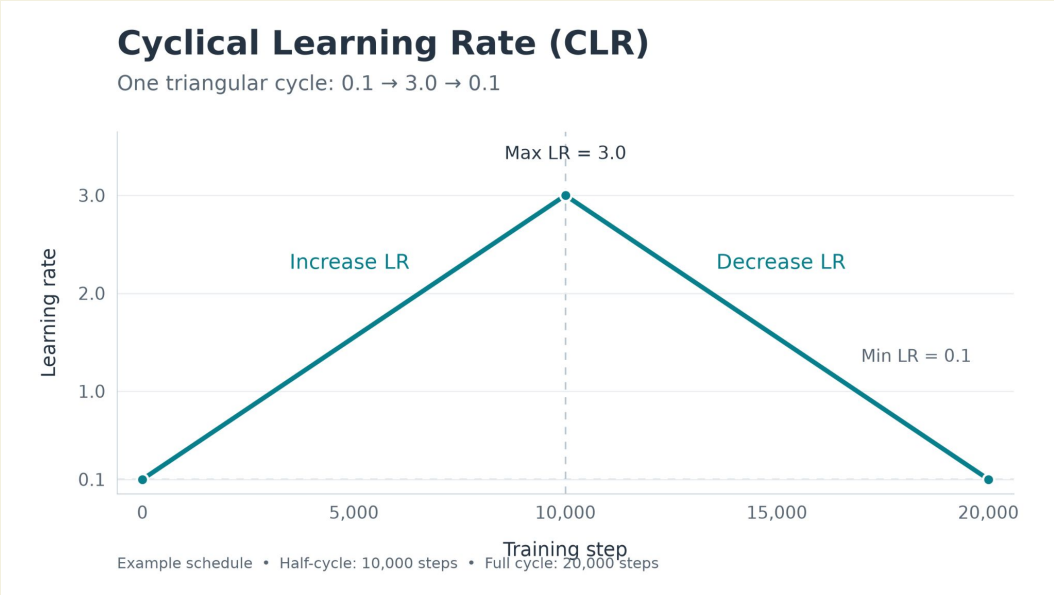
The learning rate at this peak is the largest value to use as the maximum learning rate bound when using **cyclical learning rates**. The minimum learning rate can be chosen by dividing the maximum by a factor of 3 or 4.



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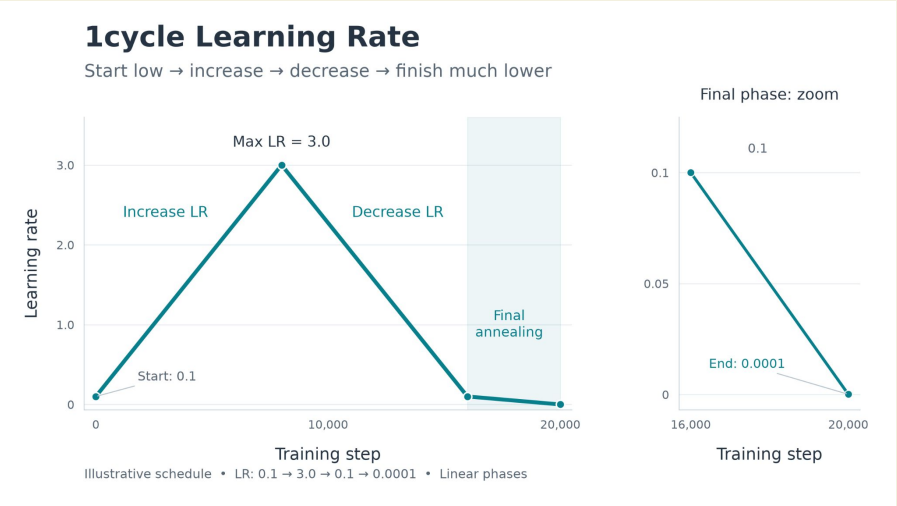
What is cyclical learning rates (CLR)

To use CLR, one specifies minimum and maximum learning rate boundaries and a stepsize. The stepsize is the number of iterations used for each step and a cycle consists of two such steps – one in which the learning rate increases and the other in which it decreases.



1cycle Learning rate

1cycle learning rate is a CLR which has final annealing stage besides basic triangle shape.



Extreme large LR can work as regularizer

When large LR appears we need to consider about making dropout and weight decay weaker than with the standard lr

Large LR

↓

Stronger regularization

↓

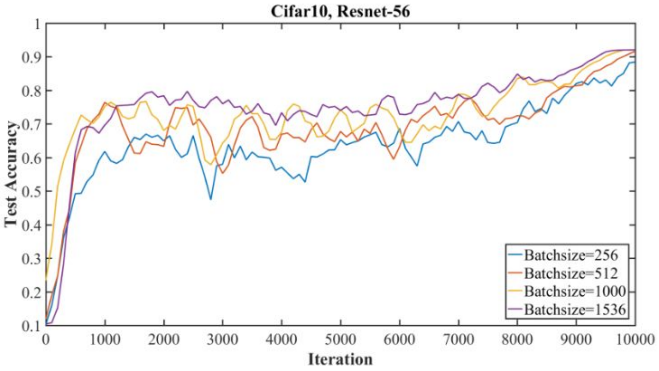
Reduce other regularization

↓

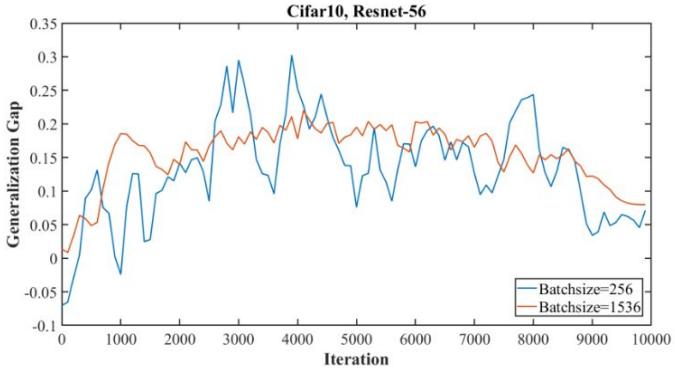
Preserve the balance

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Best batch size experiment



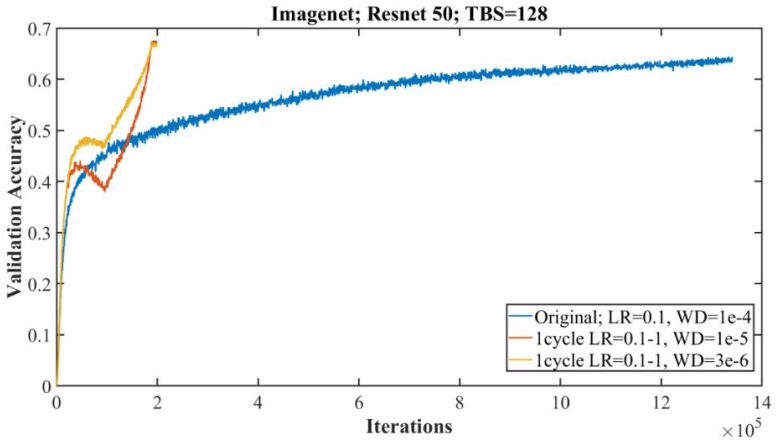
(a) Comparison of test accuracies for Cifar-10, Resnet-56 for various total batch sizes.



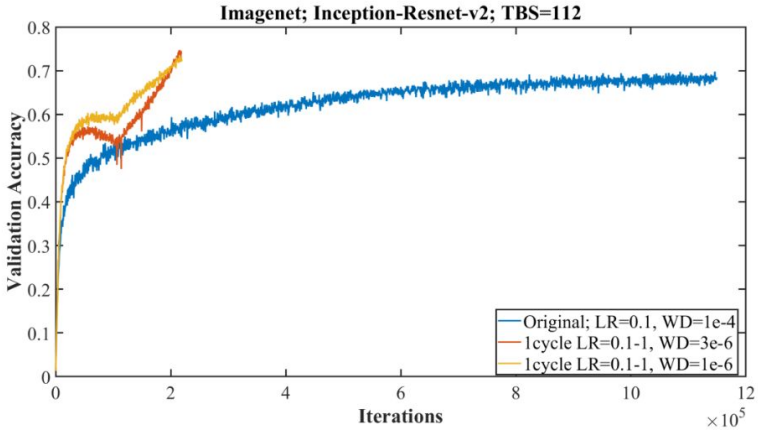
(b) Comparison of the generalization gap (training - test accuracy) for a small and a large batch size.

Figure 5: Comparisons of super-convergence to over a range of batch sizes. These results show that a large batch size is more effective than a small batch size for super-convergence training.

Imagenet task experiment



(a) Resnet-50



(b) Inception-resnet-v2

Figure 6: Training resnet and inception architectures on the imagenet dataset with the standard learning rate policy (blue curve) versus a 1cycle policy that displays super-convergence. Illustrates that deep neural networks can be trained much faster (20 versus 100 epochs) than by using the standard training methods.

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In super-convergence, networks are trained with large learning rates in an order of magnitude fewer iterations and to a higher final test accuracy than when using a piecewise constant training regime.